Top Down Dungeon Crawler Design

Cells as Player & Primary Currency

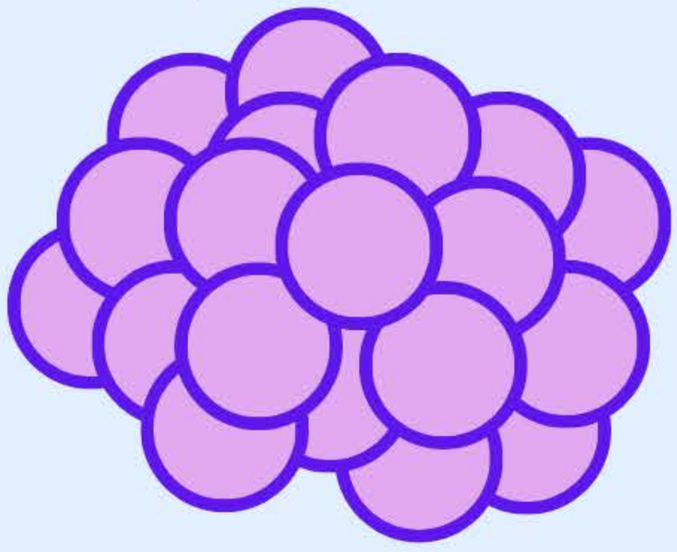
Cell Mechanics

Cells are the player's health, when the player runs out of cells they lose and must start anew.
The player gains cells from defeated enemies. The player loses cells when hit proportional to the strength of the attack they recieve.
The only upper limit on cell capacity is enemies in the environment.

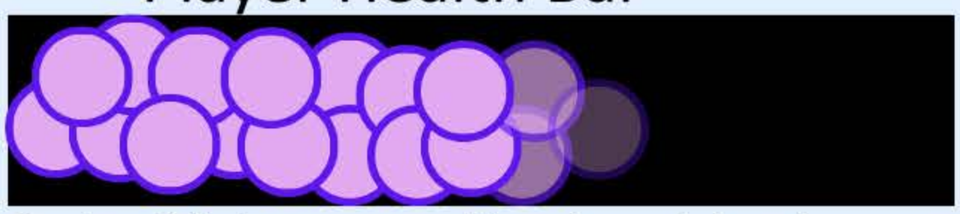
Cell Relationships

Because cells are the only visible resource, it ensures that any gain/loss of cells is important.
The player can put any almost any amount of cells in the incubator to mutate, they will always have somewhere to sink cells

Player Character

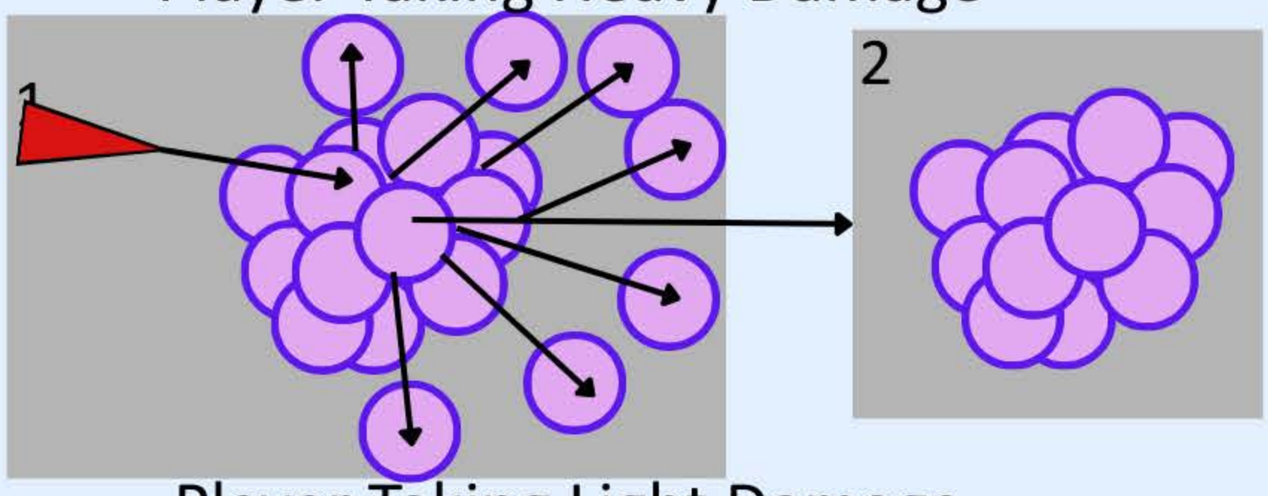


Player Health Bar

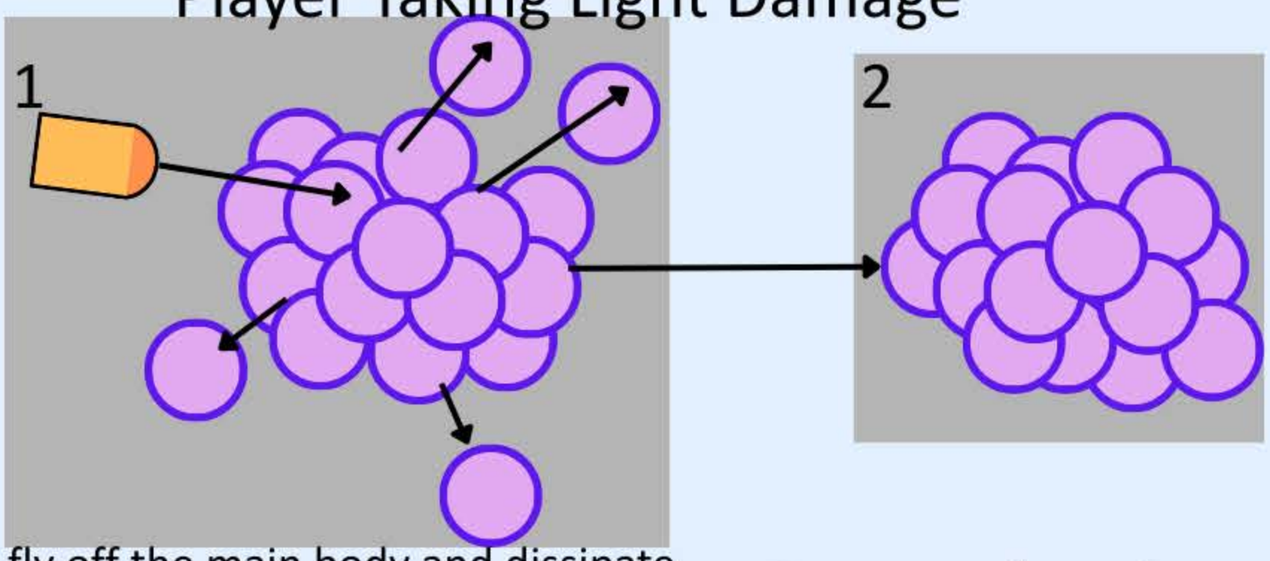


Each cell fades away as the player takes damage

Player Taking Heavy Damage



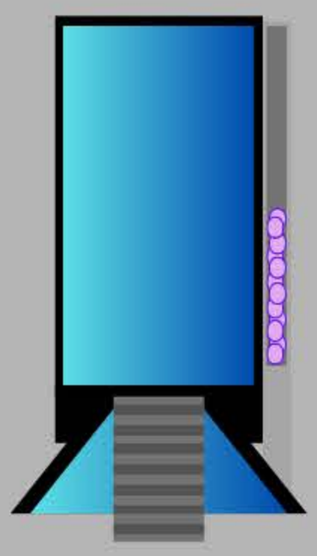
Player Taking Light Damage



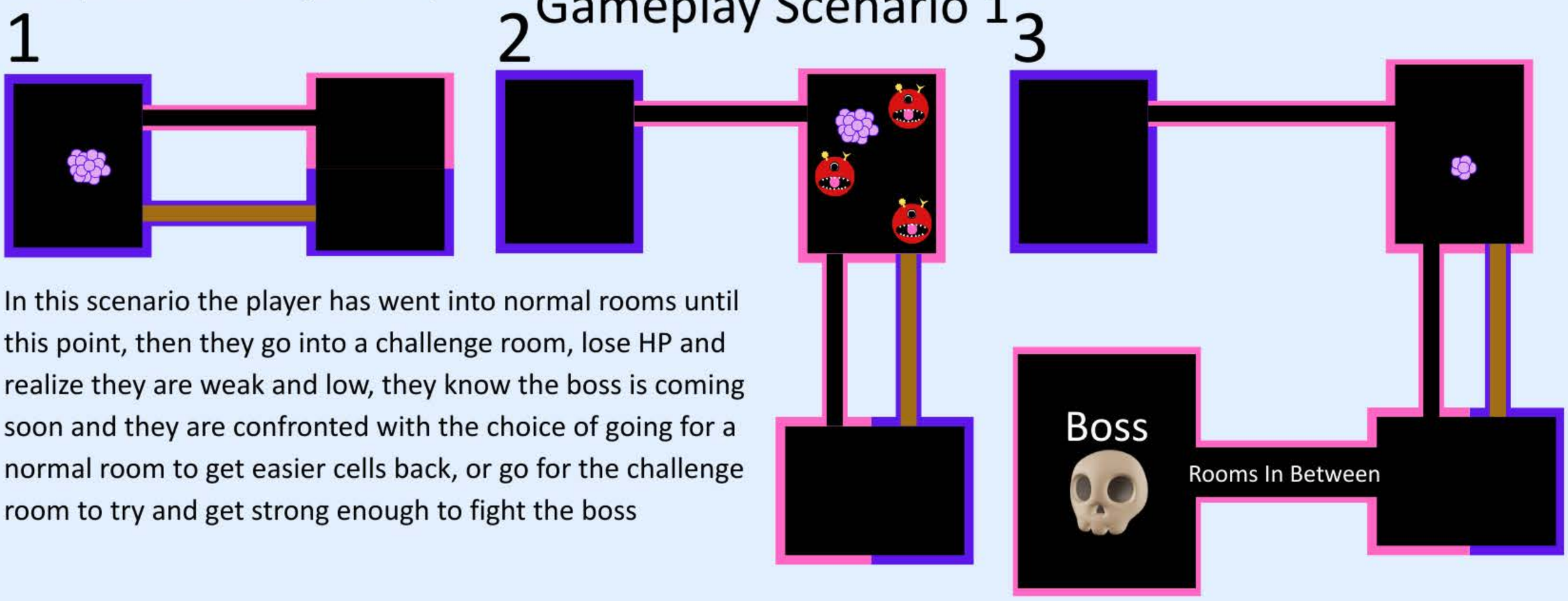
Cells fly off the main body and dissipate

Incubators

Incubators can only be found in challenge rooms and allow the player to exchange their cells for mutations (passive abilities)



Gameplay Scenario 1



In this scenario the player has went into normal rooms until this point, then they go into a challenge room, lose HP and realize they are weak and low, they know the boss is coming soon and they are confronted with the choice of going for a normal room to get easier cells back, or go for the challenge room to try and get strong enough to fight the boss

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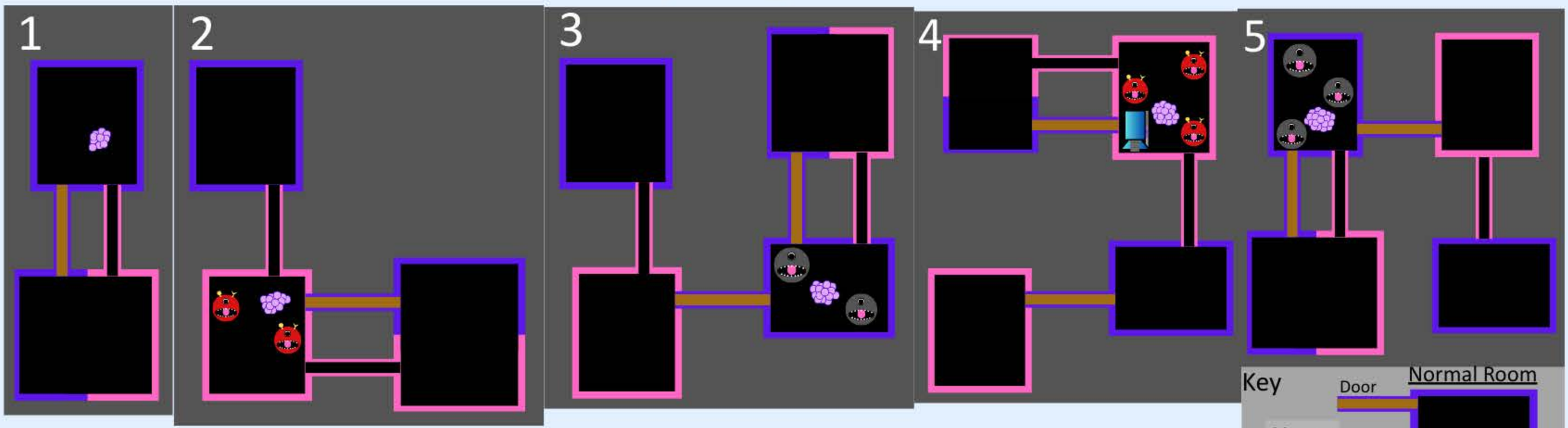
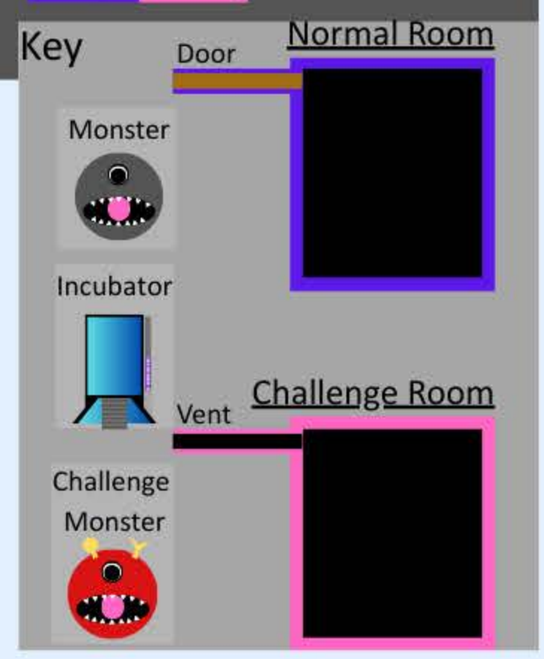
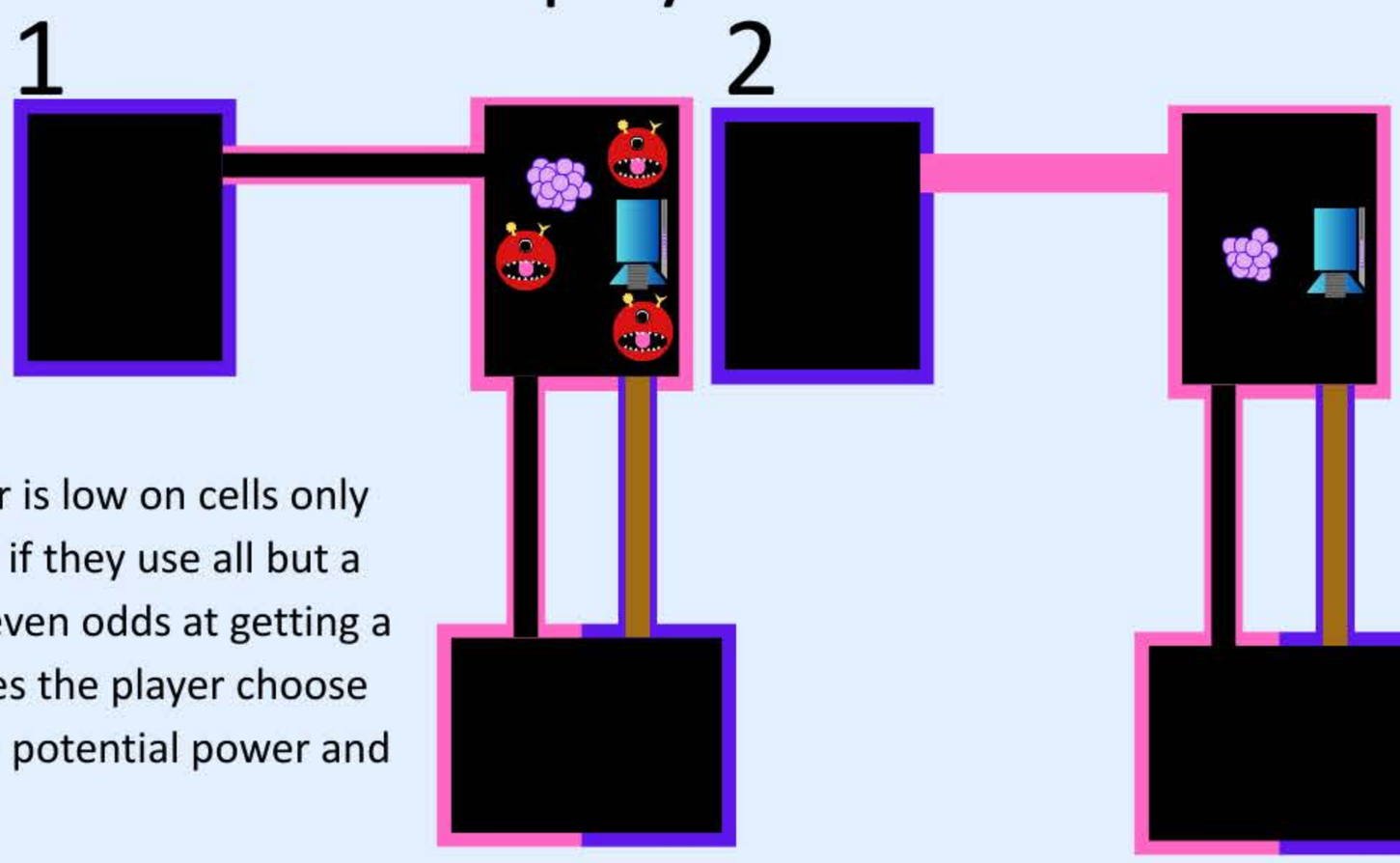
Rooms & Path Choice

Dissapearing Rooms





Gameplay Scenario 2



In this scenario the player is low on cells only 10 cells, but with 11 cells if they use all but a few cells, they still have even odds at getting a good mutation, this makes the player choose between risking it for the potential power and survival

Normal Vs Challenge Rooms

Room Mechanics:

- Whenever a player enters the next room, the path they did not take is made inactive.
- Whenever a player enter's a new room, the 3rd room behind them dissapears to make space for the next room.
- After 10 rooms the player must fight a challenging boss, too difficult for most players to beat without powers from challenge rooms.

Regular Rooms: Doors lead to regular rooms with some enemies with simple attack patterns.
Challenge Rooms: Vents lead to challenge rooms with more enemies with some complex attack patterns. Challenge rooms may contain Incubators; the only way to get stronger.

Regular Rooms vs Challenge Room Relationship:
As the player always has the option to go to the challenge room, and the challenge rooms provide better rewards, it is always optimal to enter the challenge room over the regular room, however player skill balances this out. As the player learns and gets better at the game they will choose to take on more and more challenge rooms, until they can beat that area's boss, then the challenge rooms in the next area will be too hard and they'll start the cycle again.
Normal rooms also serve as an easier source of cells if the player gets low and is concerned about dying.

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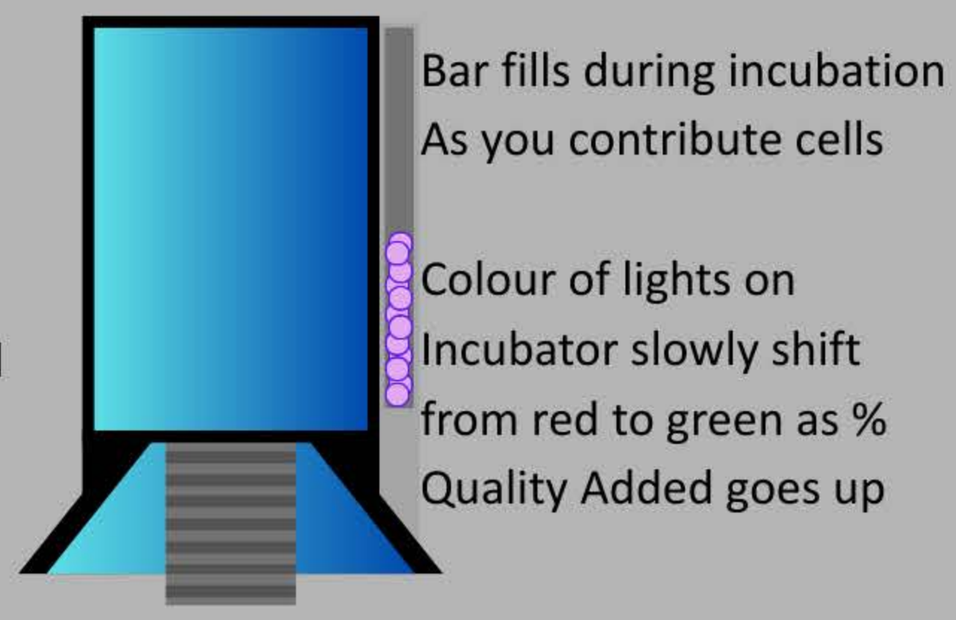
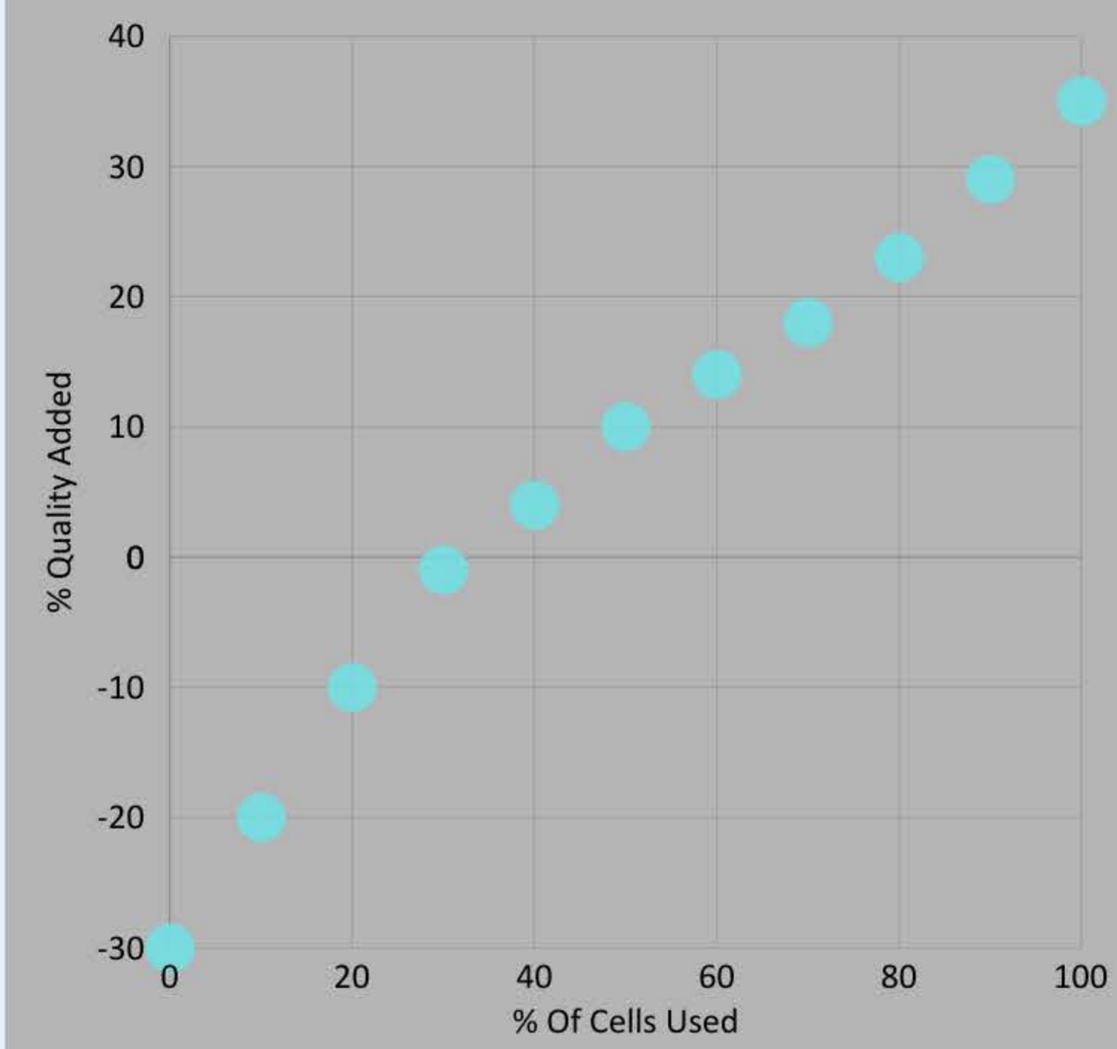
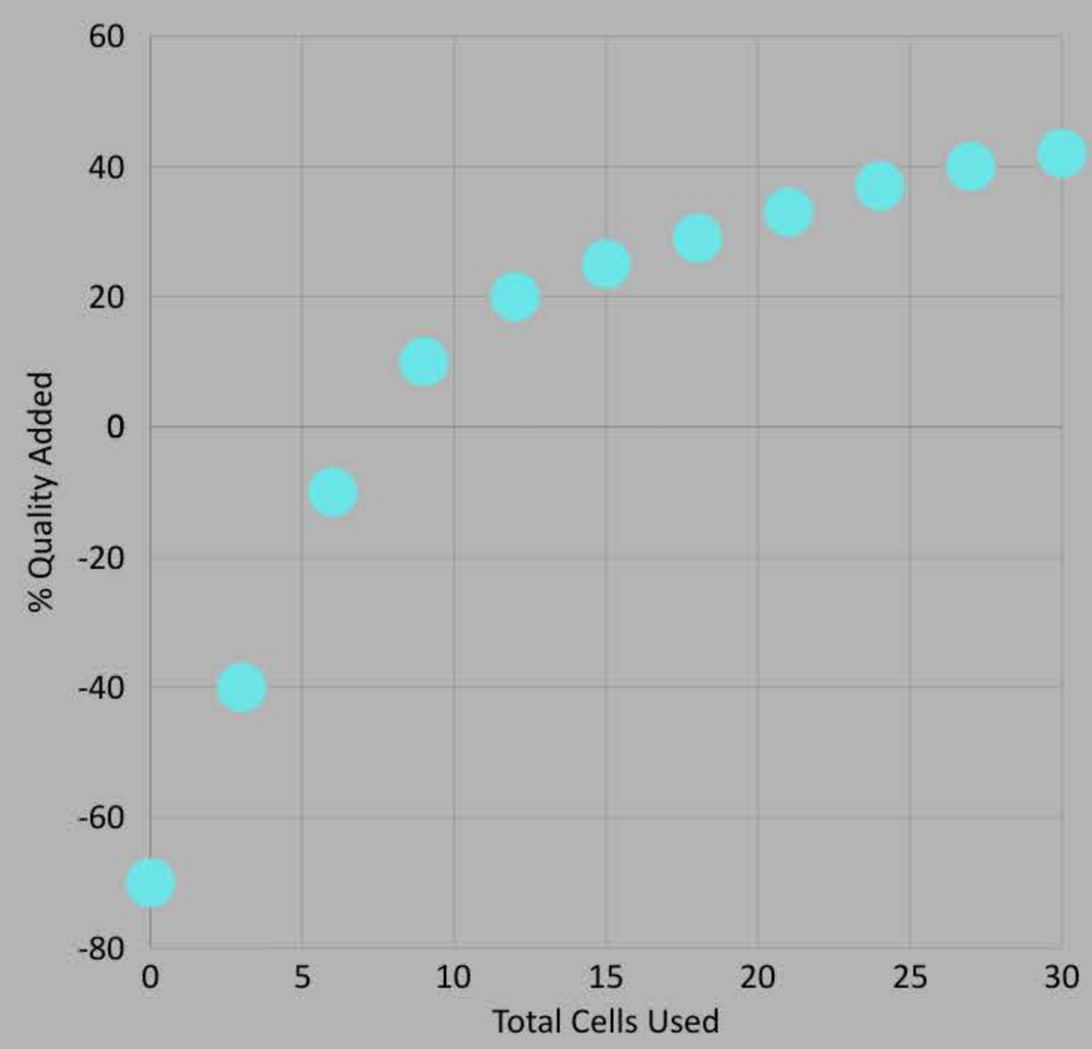
Incubator Mechanics

Incubators allow the player to spend any number of cells aslong as they have one afterwards.
The Incubator will give the player a mutation of semi-random quality determined by a loot table based on how many cells and what % of their total cells they used. This would include only negative consequences if the player tries to mutate with but a few cells.







Mutations

Each mutation has an internal Quality %, For each % Quality Added by contributing cells to the incubator the range of possible mutations goes up, when incubating the player can roll mutations with quality of within 40% + or - of their % Quality Added. Using internal Quality metrics that the player never sees allows the player to form their own beliefs of what is good.

<u>Power Mutations:</u>
Gives the player new attacks or abilities for example a Power mutation could let you throw yourr own cells as explosives. The player's default attack is throwing cells as if they are dodge balls (retrievable).

<u>Passive Mutations:</u>
Gives the player stat boosts or passively controlled abilities for example a Passive mutation could give the player 2x invincibility frame duration

<u>Maltations:</u>
Players recieve Maltations upon incubating with too few cells and rolling poorly on, Maltations give challenging effects or stat re-distribtions, for example disabling all normal rooms, or giving the player 2x cell growth but 3x damage taken.

Mechanic Rationale

<u>Cells (life as currency):</u>

<u>Object:</u>

- Each cell comprising the player

<u>Properties:</u>

- The amount of total cells comprising the player
- The colour of each cell
- The sprite for the cells

<u>Behaviours:</u>

- Moves when the player moves as it is a part of the player
- Has a chance of being destroyed when the player is hit (A certain number of random cells are destroyed when the player is hit based on the attack's damage)

<u>Relationships:</u>

- Cells are used by the rooms as the player's source of life, the rooms' inhabitants deal damage to the player destroying the cells
- Cells are used by the incubator to mutate the player (based on the number of cells used)

<u>Vent</u>

<u>Object:</u>

- The vents that lead from finished room to challenge room

<u>Properties:</u>

- The room that the vent leads to
- How many rooms back from where the current player is (tracked to know when to disable the room)
- Whether the vent is enabled

<u>Behaviours:</u>

- Transferring the player between finished rooms and challenge rooms

<u>Relationships:</u>

- Moves player to challenge room when the interact with the vent

Challenge Room

Object:

- A room with additional enemies and possibly an incubator

Properties:

- How many rooms back from where the current player is (tracked to know when to disable the room)
- Whether the room is enabled
- If there is an incubator in the challenge room and where

Behaviours:

- Spawns enemies inside for players
- Has a % chance to spawn an incubator in the room

Relationships:

- Spawns enemies when player enters the linked vent

Incubators:

Objects:

- A pod like station where the player can enter and contribute cells to gain a mutation

Properties:

- Whether the Incubator has been used yet
- How many cells the player is currently contributing

Behaviours:

- Gives the player a mutation once in exchange for their cells

Internal Relationships:

- The player receives a proportionally good mutation to both the amount of total cells the contribute to the incubation and the %of the cells they have that they contribute